DisasterAI — Retrieval Evaluation

Advanced Deep Learning — Final Project · Esprit 5DS, 2026–2027 · Group project

Section chunking (one chunk per knowledge-base entry) beats fixed 300-word chunks on every query set. On 18 blind queries, it puts the right entry first 78% of the time against 44%, with an MRR of 0.81 against 0.59.

Setup

We compared two chunking strategies on the same knowledge base, embedding model and vector store. Only the chunking changed.

- **Knowledge base:** 91 guidance entries (about 12,700 words) in 6 Markdown files: fire, flood and structural, each in English and French. Every entry has an ID (FI-xx, FL-xx, ST-xx), its source and its page.
- **Fixed chunking:** 300 words with a 40-word overlap, which gives about 50 chunks, often holding several entries.
- **Section chunking:** one chunk per entry (89 chunks). Entries over 300 words are split with the same overlap, and fragments under 30 words are merged.
- **Embeddings and store:** `intfloat/multilingual-e5-small` with its `query:` / `passage:` prefixes, and ChromaDB with cosine similarity.
- **Retrieval:** top-5, with and without the topic filter that the workflow sends (`fire`, `flood`, `structural`).

Query set	Queries	Written	Used for
v1 (`queries.csv`)	20 (12 en, 8 fr)	With the entries in view, so optimistic	First comparison
v2 (`queries_v2.csv`)	18 (10 en, 8 fr)	From user needs and the fixed incidents; committed before any run	**Main, blind comparison**
v3 (`queries_v3.csv`)	24 (6 en, 18 fr)	Before the French entries were added	French coverage
v4 (`queries_v4.csv`)	19, held out	Before the language-balance fix, with its pass rule	Testing that fix

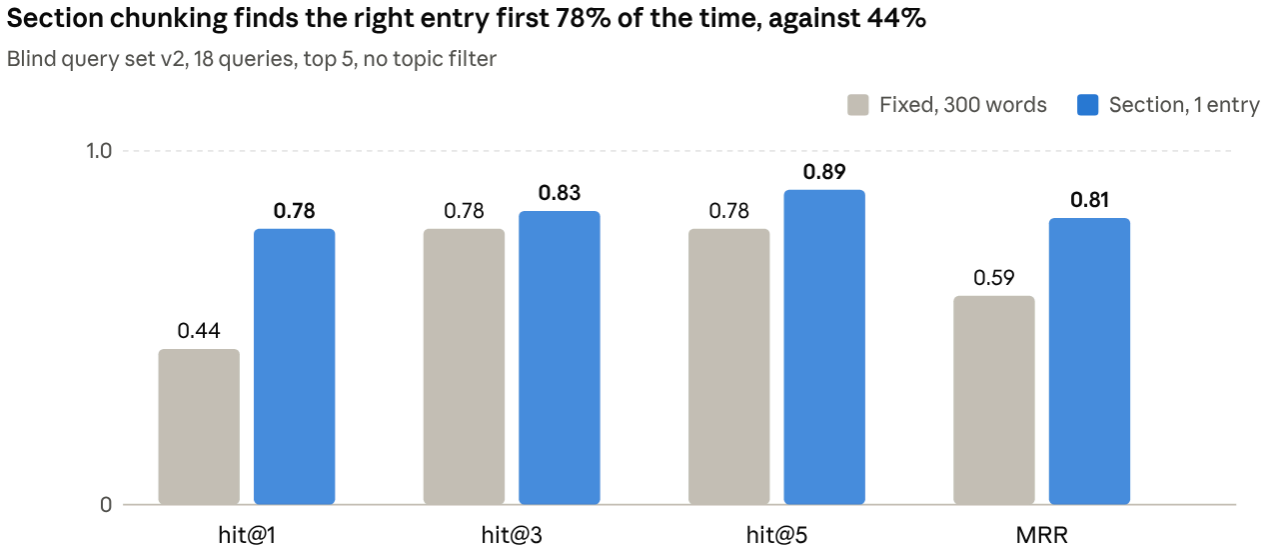
Metrics. A query is a hit at rank k when one of its expected entry IDs is in a top-k chunk.

- **hit@k** is the share of queries with a hit in the top k.

- **MRR** is the mean of 1 / rank of the first hit (0 when there is none).

Matching on entry IDs is lenient towards fixed chunks: a chunk holding three entries counts for all three. So the gap measured below, if anything, understates section chunking's advantage. Command: `python -m rag.evaluate --queries rag/eval/queries_v2.csv --strategy both [--use-topic-filter]`.

Results



rag/eval/results_queries_v2_*_20261016-1524.csv · 18 queries, 2026-10-16

The gap is largest at rank 1, where it matters most: the Knowledge agent cites what ranks first. With the topic filter used by the workflow, section chunking reaches hit@5 0.94 and MRR 0.82, against 0.89 and 0.70 for fixed. On the earlier, non-blind set v1 (20 queries), section chunking scores hit@1 0.90 and MRR 0.95, against 0.55 and 0.71.

Per query (blind set v2, no topic filter)

Rank of the first correct entry in the top 5, or "miss" when there is none. Section chunking ranks the right entry higher on 7 queries, the same on 9, and lower on 2 (v02, v15).

ID	Lang	Topic	Query	Fixed	Section
v01	en	fire	residents see flames close to the first houses and the wind pushes the fire toward them	2	1
v02	fr	fire	le feu avance vers les maisons du village, faut-il rester à l'intérieur ou partir	1	5
v03	en	fire	what to put in a go-bag and which routes to know before a wildfire	2	1

ID	Lang	Topic	Query	Fixed	Section
v04	fr	fire	protéger sa maison avec une zone dégagée de feuilles et de broussailles autour	1	1
v05	en	fire	after the blaze is out, can people go back and walk on the burned ground	1	1
v06	fr	flood	l'eau monte vite et les routes sont coupées dans mon quartier, où aller	1	1
v07	en	flood	is tap water and refrigerated food safe to consume after the flood waters go down	1	1
v08	fr	flood	combien de temps pour sécher une maison inondée avant l'apparition de moisissures	miss	miss
v09	fr	flood	danger du monoxyde de carbone avec un groupe électrogène après une inondation	3	1
v10	en	flood	what should an emergency kit for three days contain	2	1
v11	en	flood	what protective clothing for cleaning mud out of a flooded house	3	1
v12	fr	flood	puis-je rentrer chez moi après l'évacuation, des câbles électriques sont tombés	miss	1
v13	en	structural	an apartment block partly collapsed and people may be trapped, should neighbours go inside	miss	miss
v14	en	structural	how wide should the safety perimeter be around a damaged tall building	1	1
v15	en	structural	who is allowed to declare a damaged building safe or unsafe	2	3
v16	fr	structural	que signifient les affiches rouge, jaune et vert sur un immeuble endommagé	miss	1
v17	en	fire	wildfire smoke and elderly people with lung disease	1	1
v18	fr	fire	que recommande la protection civile tunisienne pour éviter les départs de feu	1	1

Raw files: `rag/eval/results_queries_v2_fixed_20261016-1524.csv` and `rag/eval/results_queries_v2_section_20261016-1524.csv`.

Comments

Why section chunking wins. A fixed chunk spans two or three entries on different subjects, so its embedding is an average of them. A query about one subject then ranks a single-subject chunk higher. With one entry per chunk, the top result is also the passage the Knowledge agent should cite. That is why we chose section chunking as the default.

The topic filter helps. Restricting the search to the incident's topic raised hit@5 on the blind set from 0.89 to 0.94 for section chunking, and from 0.78 to 0.89 for fixed. The workflow therefore always sends the topic from the Incident Understanding agent.

Misses and what we did.

- **v13 ("should neighbours go inside a collapsed block")** misses with both strategies. The right entry (ST-05, "do not enter obviously unsafe buildings") shares no words with the query, and cordon entries win instead. This is the clearest remaining weakness: the RAG agent's reworded queries are meant to cover it, but this was not measured on this case.
- **v08 (drying a flooded house)** missed because French queries were pulled towards French entries, while the answer existed only in English. We fixed this with language balancing: at least 2 results per language. On 19 held-out queries, with the pass rule written before the run, hit@5 rose from 0.42 to 0.84. The recovered answers come at ranks 4 and 5, so hit@1 did not change.
- **v02 and v15** rank lower with section chunking (5 and 3). Neighbouring entries on the same subject rank first: after-fire entries for v02, and other building-safety entries for v15. All of them still reach the 8 passages given to the Knowledge agent.

Limits. The sets are small, so one query is 5.6 points on v2. Expected entries were set by one reader. 19 French entries are team translations that no outside reader has reviewed. There is no reranker; with one entry per chunk and a topic filter, we did not find a case where one would change the 8 passages passed on.